

Mario Casas Donjuan

Monterrey, N.L., Mexico

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PROFESSIONAL SUMMARY

B.S. in Robotics Engineering undergraduate with a strong foundation in OOP (Java/Python/C++), control algorithms, and system architecture. Proven track record in competitive robotics, leading technical strategy for international competitions, gaining experience working under strict time constraints. Skilled in DevOps basics (Linux, Tunneling, Containers) and CAD design. Passionate about bridging the gap between software and hardware to create robust autonomous systems.

EDUCATION

Tecnológico de Monterrey

Monterrey, NL

B.S. in Robotics and Digital Systems Engineering (IRS)

Aug. 2025 – Jun. 2029 (Exp.)

GPA: 3.8/4.0 (Preliminary)

Prepa Tec Eugenio Garza Lagüera

Monterrey, NL

Multicultural Baccalaureate (IB Program)

Aug. 2022 – Jun. 2025

GPA: 3.6

Academic Distinctions: Honors Program, Global Citizenship Stole, LiFE STEAM Stole.

Focus on Object-Oriented Programming (Java, Python, C++) and Global Competencies.

UANL School of Music (FAMUS)

Monterrey, NL

Music Theory and Instrumental Execution (Cello)

2017 – 2022

5-year program focused on Solfège, Harmony, and Technical Execution.

TECHNICAL EXPERIENCE

FIRST Robotics Competition - Team Voltec 6647

Monterrey, NL

Team Leader & Main Driver

Aug. 2022 – Jun. 2025

Developed Object-Oriented Java code for autonomous and teleoperated control of 60 kg industrial robots.

Implemented PID control algorithms and computer vision for automatic positioning.

Proposed and implemented cross-disciplinary solutions integrating software, electronics, and mechanical subsystems.

Achievements: Regional Champion, multiple technical awards, and World Championship qualification for three consecutive years.

Escuela Voltec

Monterrey, NL

Java Programming Mentor

2024

Guided students in learning OOP and structured coding students practices with personalized exercises.

Designed course material and delivered a structured program for middle school students, helping them understand programming logic and connect the code with physical systems (robots).

PROJECTS

Home Lab Server & Portfolio | *Raspberry Pi, Docker, Nginx, React*

Jan. 2025 – Present

- Deployed a headless Linux server (Raspberry Pi OS) to host 24/7 personal services.
- Implemented Cloudflare Tunnel for secure public access (HTTPS) without exposing local ports.
- Developed a personal portfolio website using React + Vite with automated Git deployment.

FIRST Robotics Competition (2025 Season) | *2nd Place Regional & Worlds Qualifier*

Jan. 2025 – Apr. 2025

- Led the team as Captain and Main Driver, overseeing mechanical design, electronics, and programming.
- Developed structured Java code for robot control, focusing on deep debugging and system reliability.

Pharmaceutical Management App (IB Project) | *Java, MySQL, Swing*

Aug. 2024 – May 2025

- Engineered a desktop application for pharmaceutical inventory management.
- Designed a normalized relational database in MySQL and a custom authentication system.

FIRST Robotics Competition (2024 Season) | *Regional Winner & Worlds Qualifier*

Jan. 2024 – Apr. 2024

- Focused on Java programming for algorithmic optimization and mechanical integration.
- Applied algorithmic thinking to solve complex field navigation challenges.

FIRST Tech Challenge (2023 Season) | *2nd Place Regional & Top 10 National*

Dec. 2022 – Mar. 2023

- Served as Robot Driver while contributing to mechanical design and match strategy.

LEADERSHIP, DEBATE & MUN

Model United Nations (MUN) <i>Delegate - MUNMX</i> Represented a position in formal debate simulations Developed argumentation, negotiation, and public speaking skills	Monterrey,NL <i>Nov - 2023</i>
Model United Nations (MUN) <i>UNODC Committee Chair – MUNMX</i> Moderated formal debate sessions and enforced parliamentary procedure Evaluated delegates’ performance Managed crisis scenarios and guided committee workflow	Monterrey,NL <i>Nov 2024</i>

MUSICAL & ARTISTIC TRAJECTORY

FAMUS Symphony Orchestra <i>Principal Cello</i> Led the cello section in orchestral performances (Section Leader). Developed discipline, synchronous teamwork, and artistic sensitivity.	UANL <i>2017 – 2022</i>
FAMUS Guitar Ensemble <i>Principal Guitar & Soloist</i> Winner of the Faculty Guitar Contest in 2021 and 2022. Interpretation of classical and contemporary repertoire.	UANL <i>2021 – 2022</i>

SKILLS & LANGUAGES

Languages: Spanish (Native), English (C1 - Advanced), German (A1), French (A2). Programming: Java (Advanced), Python (Intermediate), C++(Intermediate), C (Beginner), TypeScript, SQL. Technologies: React, Docker, Git/GitHub, Linux (Bash/Zsh), Cloudflare, Vite. Software: VS Code, Microsoft Office, Overleaf (LaTeX), Onshape (CAD). Soft Skills: Leadership, Problem-solving, Fast learning, Work under pressure, Analytical thinking. Technical Skills: Microcontroller and system programming, automation and robotics, algorithm and data structures development	
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AWARDS & COMPETITIONS

FRC Regional Hermosillo 2024: Regional Winner (1st Place). FIRST Robotics Competitions (2022–2025): Participated in 16 regional, national, and international competitions across Mexico and the United States, including World Championships. RoboChamps 2024: Semifinalist (Advanced Category). Sports: Table Tennis Representative Team (Prepa Tec, 2022-2025).	
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